CSPCC3005: Computer Network

Teaching Scheme		Examination Scheme	
Lectures (Hours/Week)	3	ISE I (Marks)	15
Tutorials (Hours/Week)	0	ISE II (Marks)	15
Credits	3	ISE III (Marks)	10
		ESE (Marks)	60

Prerequisite
Basics of Computer & Information Technology

Course description
This course introduces the architecture, functions, components, and models of computer networks and the Internet. The principles of IP addressing and fundamentals of Ethernet concepts, media, and operations are introduced. Students will be able to build simple LANs, perform basic configurations for routers and switches, and implement IP addressing schemes.

Course Outcomes
After successful completion of the course, students will be able to:

- **CO1**: Various protocols, models in Networks.
- **CO2**: Comprehend Network hardware, Media Types (cables, Wireless).
- **CO3**: Compare UTP, Connectors, and Network Interface Card.
- **CO4**: Design, implement, and analyse simple computer networks.
- **CO5**: Apply the different strategies of TCP/UDP, FTP, HTTP, SMTP, and SNMP.

Detailed Syllabus

UNIT I	**Introduction to Network** Overview of computer network, Network hardware and software, Reference model- OSI and TCP/IP and their comparison, Network layer- Network layer design issues, various routing Algorithms and congestion control algorithms
UNIT II	**TCP/IP Architecture** TCP/IP architecture, the internet protocols, IPv4, IPv6, DHCP, and Mobile IP, IP addressing, OSPF and BGP, multicast routing, and the network layer in ATM networks
UNIT III	**Transport layer** The transport services, elements of transport protocols, internet, Transport protocols, ATM –AAL layer protocols, Performance issues.
UNIT IV	**The Application layer** Network security – principle of cryptography, secret key and public key algorithm, digital signature, Domain name system-The DNS name space, resource records, name server, Simple Network Management Protocol –SNMP model, electronic mail- architecture and services, Message formats and message transfer, email privacy.
UNIT V	**Multimedia Information and Networking** Lossless data compression, Compression of Analog Signals, Video on Demand, Image and Video Coding, ATM Layer, Transmission in ATM network, Communication satellites.

List of the Experiments

#	Title of the Experiments	Skill Level	CO	ISE Marks
1	Introduction to Networking Devices	S1	CO1	4
2	Understanding/illustrating the network features of to peer-to-peer network	S2	CO1	4
3	Understanding / Illustrating the network features of a Client-Server network.	S2	CO1	5
4	Build a Category 5 or Category 6 Unshielded Twisted Pair (UTP) Ethernet crossover cable	S2	CO3	5
5	Connecting 2 Computers using a Crossover cable	S2	CO3	5
6	Configure TCP/IP in the LAN	S3	CO5	5
7	File Transfer / Sharing/ Virtual Desktop Access	S2	CO3	5
8	Study of basic network commands and Network configuration commands.	S1	CO1	5
9	A program for a simple RSA algorithm to encrypt and decrypt the data.	S2	CO2	6
10	Client/Server chat application	S3	CO5	6

CSPCC3006: Principles of Compiler Design

Teaching Scheme		Examination Scheme	
Lectures (Hours/Week)	3	ISE I (Marks)	15
Tutorials (Hours/Week)	0	ISE II (Marks)	15
Credits	3	ISE III (Marks)	10
		ESE (Marks)	60

Prerequisite
Discrete Mathematical Structure

Course description
This course gives an introduction to system programs and compiler construction. It also gives the knowledge role of a lexical analyser, specification and recognition tokens, Lexical analyser generator LEX, role of a Parser, Types of Parsers. This course also gives an idea about Syntax Directed Translation and Intermediate Code Generation using different techniques such as DAG, three-address code, etc. At the end of this course gives information runtime environment and issues in code generation and code optimization.

Course Outcomes
After successful completion of the course, students will be able to:

- CO1: Enrich the knowledge in various phases of the compiler and its use.
- CO2: Describe the use of a lexical & syntax analyser for the generation of tokens & parse trees.
- CO3: Develop program constructs using SSDDs& type checkers
- CO4: Illustrate and demonstrate code generation techniques for targeted code
- CO5: Design a simple compiler with tools & different optimized techniques

Detailed Syllabus

UNIT I	**Introduction to Compiling and Lexical Analysis** System software, Types of System software, Definition, analysis of the source program, the phases of a compiler, the grouping of phases, Compiler Construction tools, A simple one-pass compiler, the role of the Lexical analyser, Input buffering, Specification of Tokens, Regular expressions, A Language for Specifying Lexical Analysers, Lexical Analyzer generator. (Lex)
UNIT II	**Syntax Analysis** The role of the Parser, Context-free grammars, writing a Grammar, Top-Down Parsing, Bottom-Up Parsing, Operator-precedence Parsing, LR-Parsers, Using Ambiguous Grammars, Parser Generators (YACC)
UNIT III	**Syntax-Directed Translation** Definitions, Construction of Syntax Trees, Bottom-Up Evaluation of Attributed Definitions, Top-Down Translation, Bottom-Up Evaluation of Inherited S Attributes. Type Checkers: type checking for expressions, declarations (variable, type, function, recursive), statements,
UNIT IV	**Intermediate Code Generation** Intermediate forms of source programs– abstract syntax tree, polish notation, and three-address code, types of three-address statements and their implementation, syntax-directed translation into three-address code, translation of simple statements.

UNIT V	**Code Generation & Code Optimization** Issues in the Design of a Code Generator, the target Machine, Run-Time Storage Management, Basic Blocks and Flow Graphs, Simple Code Generator, register allocation and Assignment, The DAG Representation of Basic Blocks, Generating Code from DAGs, Peephole Optimization, Principal sources of optimization.

List of the Experiments

#	Title of the Experiments	Skill Leve l	CO	ISE Marks
1	Study of LEX Analyzer Generator	S1	CO1	04
2	Study of YACC Generator	S1	CO1	04
3	Write a C program to identify whether a given line is a comment or not	S1	CO1, CO2	04
4	Implement a lex program to count the no. of vowels & consonants in a given string	S1	CO3	04
5	Implement a lex program to count the no. of characters, words, spaces & end lines in a given input file	S2	CO2, CO3	04
6	Implement a lex program to count the no. of -ve & +ve integers & fractions	S2	CO2, CO3	04
7	Implement a program to find out whether the string is a keyword or not	S2	CO2	04
8	Implement a program to find out whether the string is an identifier or not	S2	CO2	04
9	Implement a program to find out whether the string is constant or not	S2	CO2	04
10	Write a C program to recognize strings under 'a*', 'a*b+', and 'abb'.	S2	CO3	04
11	Evaluate a C program for the generation of a Parse tree 64	S2	CO4	04
12	Implement the lexical analyser using JLex, flex, or other lexical analyser generating tools.	S2	CO5	06

Detailed Syllabus

UNIT I	**Digital Image Formation and low-level processing**: Image representations, a few concepts, Image digitization, Digital image properties, Colour images. Images as stochastic processes, Image formation physics. **Transformation Background**: Linear integral transforms, Orthogonal, Euclidean Image Enhancement, Restoration, Histogram Processing
UNIT II	**Image Formation:** Geometric primitives and transformation, photometric image formation, the digital camera. **Data structures for image analysis:** levels of image data representation, traditional image data structures, and Hierarchical data structures. Image understanding-fitting via random sample consensus, point distribution model.
UNIT III	**Image Segmentation:** Region Growing, Edge-Based approaches to segmentation, Object detection. Mean Shift Segmentation MRFs, Active contour models – snakes, 3D graph–based image segmentation, Graph cut segmentation. **Pattern Analysis**: Clustering: K-Means, K-Medoids, Mixture of Gaussians
UNIT IV	**3D Transformation:** Vision tasks, basics of projective geometry, A Single perspective camera, Scene reconstruction from multiple views, two camera stereopsis, 3D Reconstruction**:** Shape from X, Full 3D objects, 3D model-based vision, 2D view-based representations of a 3D scene.
UNIT V	**Motion Analysis:** Time difference, Background subtraction and modelling, Optical flow, Detection of specific motion patterns, Dynamic Stereo, video tracking, **Human Analysis:** pose estimation, facial analysis, attribute recognition.

List of the Experiments

The student shall perform a minimum of ten experiments of the following using Python.

#	Title of the Experiments	Skill Level	CO	ISE Marks
	Level: Basic (all)			
1	To implement Contours for boundaries of the shape to detect certain types of shapes.	S1	CO1	02
2	To implement the graph cut segmentation	S2	CO2	02
3	To implement the mean shift segmentation	S2	CO2	02
4	To obtain a histogram equalization image	S2	CO2	03
5	To implement a colour detection and edge detection algorithm	S2	CO2	03
	Level: Moderate (all)			
6	To implement Object tracking with particle filters	S3	CO3	02
7	To implement the hand gesture recognition	S3	CO4	02
8	To implement the 3D object	S2	CO3	02
9	To implement a smoothing or averaging filter in the spatial domain.	S3	CO4	02
	Level: Complex (any one)			
10	Implement object counting.	S3	CO3	02
11	Implement human counting.	S3	CO3	03

UNIT I	**Overview of Natural Language Processing (NLP) and its applications in AI.** *Understanding linguistic fundamentals:* syntax, semantics, morphology, and phonetics. Tokenization, stemming, and lemmatization. *Text Preprocessing and Feature Extraction:* Techniques for cleaning and preprocessing textual data, Feature extraction methods for representing text data, including bag-of-words and TF-IDF. *Introduction to popular NLP libraries:* NLTK, spaCy, Hugging Face, Transformers.
UNIT II	**Word Level Analysis:** Unsmoothed n-grams, evaluating n-grams, smoothing part-of-speech tagging, rule-based, stochastic, and transformation-based tagging, issues in pos tagging – Hidden Markov and maximum entropy models, Parsing, Regular Expressions for Text Processing
UNIT III	**Text Representation Techniques:** Bag of Words (BoW), Term Frequency-Inverse Document Frequency (TF-IDF), Word Embeddings: Word2Vec, GloVe, FastText. Sentiment analysis, named entity recognition, and part-of-speech tagging.

UNIT IV	**Text Classification:** Supervised vs. Unsupervised Learning in NLP, Naïve Bayes, Logistic Regression, and SVM for Text Classification, News categorization and summarization, topic modelling, Latent Dirichlet Allocation
UNIT V	**Introduction to Large Language Models (LLM):** Overview of large language models such as GPT (Generative Pre-trained Transformer) and BERT (Bidirectional Encoder Representations from Transformers). Pre-training Language Models: Understanding the pre-training process for language models, Exploration of model architectures, and training strategies.

#	Title of the Experiments	Skill Level	CO	ISE Marks
1	Write a Python script to read data and implement stop word removal using NLTK. Convert text to lowercase, remove punctuation, and handle contractions. Use regex to extract email IDs, phone numbers, or URLs from text.	SO1	CO1	3
2	Write a Python script to tokenize a given text using NLTK and spaCy. Perform stemming and lemmatization.	SO2	CO1	3
3	Write a Python script to implement BoW and TF-IDF on a dataset using Scikit-learn.	SO2	CO2	2
4	Write a Python script to implement an N-gram model for text generation: Extract unigrams, bigrams, and trigrams from a given text, and generate new text using an N-gram language model.	SO2	CO2	2
5	Write a Python script to train a model (Naïve Bayes, SVM) to classify emails as spam or not spam.	SO3	CO3	2
6	Write a Python script to train Word2Vec embeddings using Gensim, and Load pre-trained GloVe embeddings for text similarity analysis.	SO3	CO3	3
7	Write a Python script to train a Naïve Bayes and Logistic Regression classifier for sentiment analysis. 72	SO3	CO3	3
8	Write a Python script to implement LDA to discover hidden topics in a set of documents.	SO4	CO5	3
9	Write a Python script to build a question-answering system using a fine-tuned BERT model.	SO4	CO5	3
10	Write a Python script to implement text summarization using NLP techniques and transformers.	SO4	CO4	2

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Approved Updated Curriculum in XXth Academic C
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Level: Basic (all)				
1	Creating an Environment Setup for React	S1	CO1	04
2	Creating your first React Program	S2	CO1, CO2	04
3	Write a program to create a simple calculator Application using React JS	S2	CO2	04
4	Create a Simple Login and registration page using React JS	S2	CO2	04
5	Creating Function and Class Components	S2	CO3	04
6	Implement component inheritance by rendering a message in the Child Component passed from the Parent component.	S2	CO3	04
7	State and Lifecycle management using React	S2	CO3	04
8	Demonstrate Lists and Keys in React	S2	CO3	04
Level: Moderate (all)				
9	Create a Webpage to handle events in React	S2	CO4	06
10	Create a program using Composition in React	S2	CO4	06
11	Creating Simple Forms using React	S2	CO5	06
12	Implement the concept of conditional rendering in React	S2	CO5	06
13	Demonstrate the use of state hooks	S2	CO5	06
14	Implement a program to build your Hooks in React	S2	CO5	06
Level: Complex (all)				
15	Deploying Webpages on IIS and Demonstrating FileZilla	S2	CO4	06
16	Mini Project: Create a project on a Grocery delivery application	S2	CO3, CO5	06
17	Mini Project: Create a to-do list app using React	S2	CO3, CO5	06